Qué protege una simetría parcial

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$\mathcal{L}(r,\dot r,\dot\varphi) = \frac{1}{2}m(\dot r^2\csc^2\alpha + r^2\dot\varphi^2) - mgr\cot\alpha$

a) $x = r\,\text{sen}\,\alpha\cos\varphi$ $\dot x = \dot r\,\text{sen}\,\alpha\cos\varphi - r\,\text{sen}\,\alpha\,\text{sen}\,\varphi\cdot\dot\varphi$

$y = r\,\text{sen}\,\alpha\,\text{sen}\,\varphi$ $\dot y = \dot r\,\text{sen}\,\alpha\,\text{sen}\,\varphi + r\,\text{sen}\,\alpha\cos\varphi\cdot\dot\varphi$

$z = r\cot\alpha$ $\dot z = \dot r\cot\alpha$

$L = r\times mv$ $L = m(r\times v)$

$\begin{vmatrix} x & y & z \\ \dot x & \dot y & \dot z \end{vmatrix} = (y\dot z - z\dot y)\hat i - (x\dot z - z\dot x)\hat j + (x\dot y - y\dot x)\hat k$

$L_x = m(y\dot z - z\dot y)$; $y\dot z = (r\,\text{sen}\,\alpha\,\text{sen}\,\varphi)(\dot r\cot\alpha)$ csc → 1/sen, cot → 1/tan, sec → 1/cos

$L_y = m(x\dot z - z\dot x)$ $y\dot z = r\cdot\dot r\,\text{sen}\,\varphi\cdot\cancel{\text{sen}\,\alpha}\cdot\frac{\cos\alpha}{\cancel{\text{sen}\,\alpha}}$

$y\dot z = r\cdot\dot r\,\text{sen}\,\varphi\cdot\cos\alpha$

$z\dot y = r\cot\alpha(\dot r\,\text{sen}\,\alpha\,\text{sen}\,\varphi + r\,\text{sen}\,\alpha\cos\varphi\cdot\dot\varphi)$

$z\dot y = r\dot r\cos\alpha\,\text{sen}\,\varphi + r^2\cos\alpha\cos\varphi\cdot\dot\varphi$

$y\dot z - z\dot y = r\cdot\dot r\,\text{sen}\,\varphi\cdot\cos\alpha - r\cdot\dot r\cos\alpha\,\text{sen}\,\varphi - r^2\cos\alpha\cos\varphi\cdot\dot\varphi$

$y\dot z - z\dot y = -r^2\cos\alpha\cos\varphi\cdot\dot\varphi$

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$x\dot z = r\,\text{sen}\,\alpha\cos\varphi(\dot r\cot\alpha)$

$z\dot x = (r\cot\alpha)(\dot r\,\text{sen}\,\alpha\cos\varphi - r\,\text{sen}\,\alpha\,\text{sen}\,\varphi\cdot\dot\varphi)$

$x\dot z = r\cdot\dot r\cos\varphi\cos\alpha$

$z\dot x = r\cdot\dot r\cos\alpha\cos\varphi - r^2\cos\alpha\,\text{sen}\,\varphi\cdot\dot\varphi$

$z\dot x - x\dot z = r\cdot\dot r\cos\alpha\cos\varphi - r^2\cos\alpha\,\text{sen}\,\varphi\cdot\dot\varphi - r\cdot\dot r\cos\varphi\cos\alpha$

$= -r^2\cos\alpha\,\text{sen}\,\varphi\cdot\dot\varphi^2$

$L_x = -mr^2\cos\alpha\cos\varphi\cdot\dot\varphi$ $L_x = -mr^2\dot\varphi\cos\alpha\cos\varphi$

$L_y = -mr^2\cos\alpha\,\text{sen}\,\varphi\cdot\dot\varphi$ $L_y = -mr^2\dot\varphi\cos\alpha\,\text{sen}\,\varphi$

$\frac{\partial\mathcal L}{\partial\dot\varphi} = mr^2\dot\varphi = p_\varphi$ $L_x = p_\varphi\cos\alpha\cos\varphi$

$L_y = p_\varphi\cos\alpha\,\text{sen}\,\varphi$

¿$\frac{\partial\mathcal L}{\partial x} = 0$ $\frac{\partial\mathcal L}{\partial y} = 0$? x e y se pueden obtener a través de φ,

Lx y Ly no se conservan

b) $|L_\perp|^2 = L_x^2 + L_y^2$

$|L_\perp|^2 = p_\varphi^2\cos^2\alpha\cos^2\varphi + p_\varphi^2\cos^2\alpha\,\text{sen}^2\varphi$

$|L_\perp|^2 = p_\varphi^2\cos^2\alpha(\cos^2\varphi + \text{sen}^2\varphi)$

$|L_\perp|^2 = p_\varphi^2\cos^2\alpha$

$|L_\perp| = p_\varphi\cos\alpha$ $p_\varphi = \text{cste}$ y α → ángulo de abertura del cono, constante

$|L_\perp| = \text{cste}$

Una cantidad es conservada si es constante a lo largo de la trayectoria. Una magnitud puede ser constante sin que las componentes individuales lo sean.

Lx y Ly varían con α

c) $h = \sum_j \dot q_j\frac{\partial\mathcal L}{\partial\dot q_j} - \mathcal L$

$h = \dot r\frac{\partial\mathcal L}{\partial\dot r} + \dot\varphi\frac{\partial\mathcal L}{\partial\dot\varphi} - \mathcal L$

$\frac{\partial\mathcal L}{\partial\dot r} = m\dot r\csc^2\alpha$ $\frac{\partial\mathcal L}{\partial\dot\varphi} = mr^2\dot\varphi = p_\varphi$ $p_\varphi = m^2r^4\dot\varphi^2$ $\frac{p_\varphi}{mr^2} = mr^2\dot\varphi^2$

$h = \dot r(m\dot r\csc^2\alpha) + \dot\varphi(mr^2\dot\varphi) - \frac{1}{2}m(\dot r^2\csc^2\alpha + r^2\dot\varphi^2) + mgr\cot\alpha$

$h = \dot r^2 m\csc^2\alpha - \frac{1}{2}\dot r^2 m\csc^2\alpha + \dot\varphi^2 mr^2 - \frac{1}{2}\dot\varphi^2 mr^2 + mgr\cot\alpha$

$h = \frac{1}{2}\dot r^2 m\csc^2\alpha + \frac{1}{2}\dot\varphi^2 mr^2 + mgr\cot\alpha$

$h = \frac{1}{2}m(\dot r^2\csc^2\alpha + r^2\dot\varphi^2) + mgr\cot\alpha$

$h = \frac{1}{2}m\dot r^2\csc^2\alpha + \frac{1}{2mr^2}p_\varphi + mgr\cot\alpha$ Energía cinética radial

$\frac{1}{2}m\dot r^2\csc^2\alpha = \frac{1}{2}mv^2$

$m_{ef} = \frac{1}{2}m\dot r^2\csc^2\alpha$

$E = \frac{1}{2}m_{ef}\dot r^2 + V_{ef}(r)$

$V_{ef}(r) = E - \frac{1}{2}m_{ef}\cdot\dot r^2$

$V_{ef}(r) = \frac{1}{2}\frac{p_\varphi^2}{mr^2} + mgr\cot\alpha$

$E = \frac{1}{2}m\csc^2\alpha\cdot\dot r^2 + V_{ef}(r)$

$r_0 = r$ $\dot r = 0$ $\ddot r = 0$

$\frac{dV_{ef}(r)}{dr} = 0$ → qué pasa en los extremos

$\frac{dV}{dr} = \frac{1}{\cancel{2}}\frac{p_\varphi^2}{m}\cdot -\cancel{2}\frac{1}{r^3} + mg\cot\alpha$

$\frac{dV}{dr} = -\frac{p_\varphi^2}{mr^3} + mg\cot\alpha$

$0 = -\frac{p_\varphi^2}{mr^3} + mg\cot\alpha$

$\frac{p_\varphi^2}{mr^3} = mg\cot\alpha$

$\frac{p_\varphi^2}{m^2g}\tan\alpha = r^3$

$\left(\frac{p_\varphi^2\tan\alpha}{mg}\right)^{1/3} = r_0$

d) $m\csc^2\alpha\,\ddot r = -\frac{dV_{ef}}{dr}$ Pensemos $|\delta r| << r_0$ linealización alrededor de r_0 $r(t) = r_0 + \delta r(t)$

$V_{ef}(r_0 + \delta r) \approx V_{ef}(r_0) + \frac{dV_{ef}}{dr}\Big|_{r_0}\delta r + \frac{1}{2}\frac{d^2V_{ef}}{dr}(\delta r)^2$

$\frac{dV_{ef}}{dr}\Big|_{r_0} = 0$

$\frac{dV_{ef}}{dr}\Big|_{r_0+\delta r} \approx \frac{d^2V_{ef}}{dr^2}\Big|_{r_0}\delta r$

$m\csc^2\alpha\,\delta\ddot r = -\frac{d^2V_{ef}}{dr^2}\Big|_{r_0}\delta r$

$k = \frac{d^2V_{ef}}{dr^2} > 0$ (por estabilidad

$\delta\ddot r + \frac{k}{m\csc^2\alpha}\delta r = 0$

$\omega_r^2 = \frac{k}{m\csc^2\alpha}$

$V_{ef}(r) = \frac{p_\varphi^2}{2mr^2} + mgr\cot\alpha$

$V'_{ef}(r) = -\frac{p_\varphi^2}{mr^3} + mg\cot\alpha$

$V''_{ef}(r) = \frac{3p_\varphi^2}{mr^4}$

$r = r_0$ $p_\varphi^2 = m^2gr_0^3\cot\alpha$

$V''_{ef} = \frac{3mg\cot\alpha}{r_0}$

$\omega_r^2 = \frac{1}{m\csc^2\alpha}\cdot\frac{3mg\cot\alpha}{r_0} = \frac{3g\cot\alpha\csc^2\alpha}{r_0}$

$p_\varphi = mr_0^2\dot\varphi$ $\dot\varphi_0$ es la velocidad angular $p_\varphi^2 = m^2gr_0^3\cot\alpha$

$\dot\varphi_0^2 = \frac{g\cot\alpha}{r_0}$

$\omega_r^2 = 3\csc^2\alpha\cdot\dot\varphi_0^2$

$\omega_r = \sqrt{3}\csc\alpha\cdot\dot\varphi_0$

e) $\dot\varphi_0^2 = \frac{g\cot\alpha}{r_0}$ $\dot\varphi_0 = \sqrt{\frac{g\cot\alpha}{r_0}}$

$\frac{\omega_r}{\dot\varphi_0} = \frac{\sqrt{3}\csc\alpha\cdot\dot\varphi_0}{\dot\varphi_0}$

$\frac{\omega_r}{\dot\varphi_0} = \sqrt{3}\csc\alpha$

f) $\frac{\omega_r}{\dot\varphi_0} = \frac{p}{q}$ 1. Potencial kepleriano $V \propto -1/r$: $\omega_r/\dot\varphi = 1$

2. Oscilador ($V \propto r^2$): $\omega_r/\dot\varphi = 1$

g) $\alpha \to 0$ $\alpha \to \pi/2$